Inline and Display Mathematics

Source-backed grouped formula sample

Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\frac{\partial \mathcal{V}}{\partial \phi} = \frac{2}{n}\frac{1}{\operatorname{Tr}(\mathcal{M}^2)}\mathcal{M}_{mn}\frac{\partial \mathcal{V}}{\partial \mathcal{M}_{mn}}\,, \Rightarrow \mathcal{M}^{mn}\mathcal{M}_{pq}\frac{\partial \mathcal{V}}{\partial \mathcal{M}_{pq}} - \operatorname{Tr}(\mathcal{M}^2)\frac{\partial \mathcal{V}}{\partial \mathcal{M}_{mn}} = 0\,.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\int DV \exp\left(\frac{1}{2}\int d^4x \varepsilon^{\mu\nu\tau\lambda} + W^{\alpha\beta}_{\mu\nu} + G^{\sigma\rho}_{\tau\lambda}\varepsilon_{\alpha\beta\sigma\rho} - \frac{1}{2}\int d^4x \varepsilon^{\mu\nu\tau\lambda} - W^{\alpha\beta}_{\mu\nu} - G^{\sigma\rho}_{\tau\lambda}\varepsilon_{\alpha\beta\sigma\rho}\right).$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$I = \int_\Sigma d^3x\sqrt{\gamma}\Big(R(\gamma) - \frac{1}{2}((DU)^2 - e^{-2U}(DV)^2) - \frac{1}{2}((D\Phi)^2 - e^{-2\Phi}(D\Psi)^2) - (DT)^2 + e^{-(U+\Phi)}V(T)\Big)$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$2\frac{\int_{b=\theta_2}^{b=\theta_1} e^{\frac{\rho r s\cos(b)}{1-\rho^2}}\,db}{2\pi\int_0^{2\pi} e^{\frac{\rho r s\cos a}{1-\rho^2}}\,da} \leq 2\frac{\int_{b=0}^{b=\theta_1-\theta_2} e^{\frac{\rho r s\cos(b)}{1-\rho^2}}\,db}{2\pi\int_0^{2\pi} e^{\frac{\rho r s\cos a}{1-\rho^2}}\,da} < \frac{\theta_1-\theta_2}{2\pi^2}.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\frac{d}{d\lambda}S_{\omega,\boldsymbol{c}}\big(\Phi_0^\lambda\big) = -\Big(\sqrt{\lambda}-1\Big)(2L(\Phi_0)+\boldsymbol{c}\cdot\boldsymbol{P}(\Phi_0))\left(\lambda + \frac{\boldsymbol{c}\cdot\boldsymbol{P}(\Phi_0)}{2L(\Phi_0)+\boldsymbol{c}\cdot\boldsymbol{P}(\Phi_0)}\sqrt{\lambda} + \frac{\boldsymbol{c}\cdot\boldsymbol{P}(\Phi_0)}{2L(\Phi_0)+\boldsymbol{c}\cdot\boldsymbol{P}(\Phi_0)}\right)$$